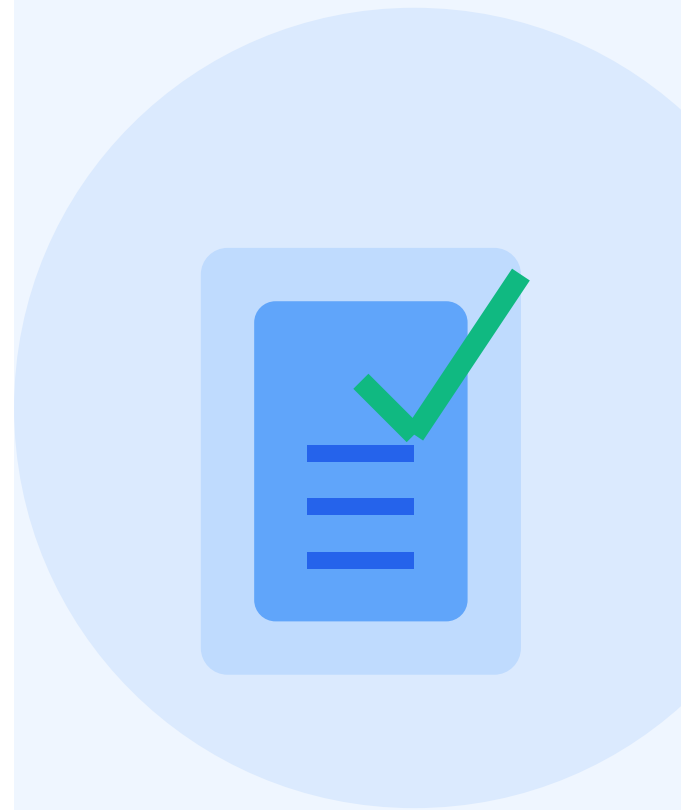


Smart Loan Risk & Approval System

Business Report

End-to-End Machine Learning Application
for Loan Risk Prediction and Approval



Report Summary

Report Type:	Business Report
Generated On:	2026-10-03 23:08
Total Records:	102
Total Columns:	21
Target Column:	Loan_Status
Active Model:	Logistic Regression

✓ Data Analysis & Insights

✓ Risk Prediction

✓ Approval Decision

✓ Better Financial Decisions

1. Dataset Overview

Total Records

102

Total Columns

21

Numeric Cols

11

Categorical Cols

10

2. Data Quality

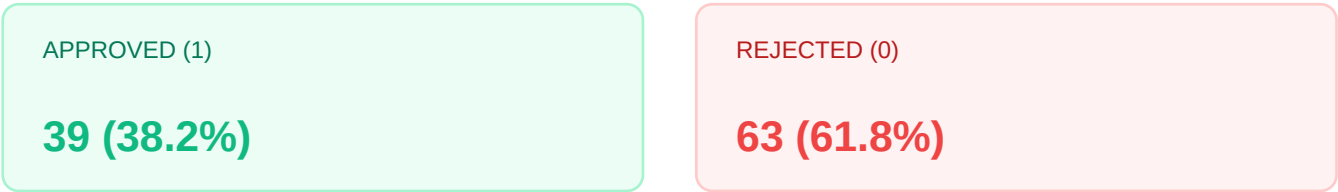
Metric	Count	Percentage
Missing Values	0	0.00%
Duplicate Rows	0	0.00%
Numerical Cols	11	52%
Categorical Cols	10	48%

✓ Dataset is ready for analysis and model training.

3. Missing Values by Column

✓ No missing values detected.
All columns have complete data.

4. Target Analysis (Loan_Status)



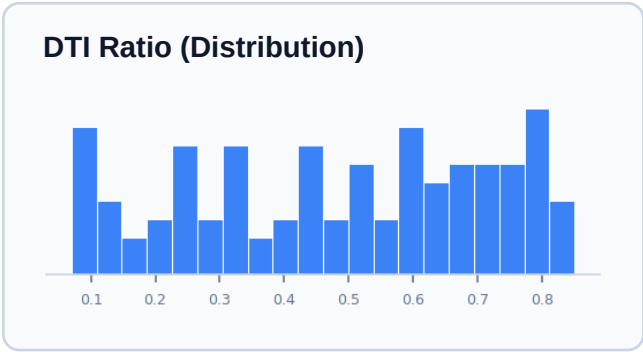
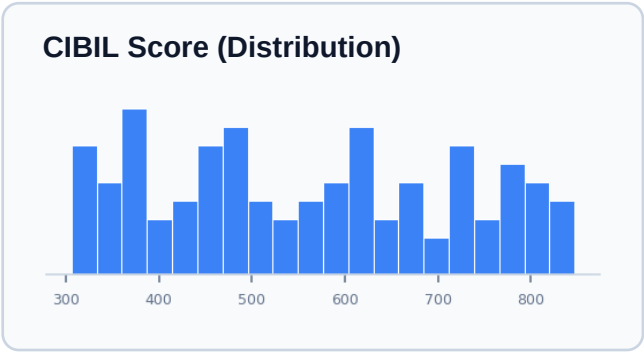
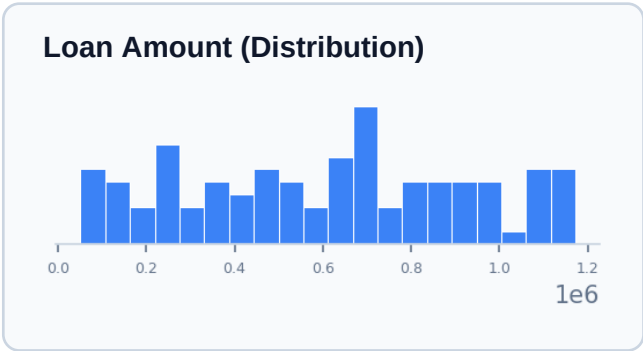
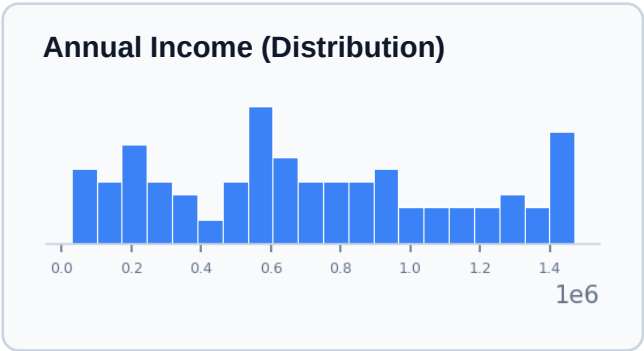
5. Descriptive Statistics

Feature	Mean	Median	Min	Max	Std Dev
Age	42.8	43.0	21.0	65.0	13.3
Employment_Years	16.1	18.0	0.0	30.0	9.1
Annual_Income	709,578.06	669,646.00	28,267.00	1,471,403.00	422,109.94
Coapplicant_Income	176,856.75	179,208.00	1,542.00	394,540.00	108,531.35
Loan_Amount	608,104.39	638,579.00	51,062.00	1,172,262.00	319,758.03
Loan_Term_Years	12.5	10.0	2.0	30.0	9.0
Cibil_Score	559.5	553.0	306.0	847.0	158.2
Existing_Loans	2.6	3.0	0.0	5.0	1.7
DTI_Ratio	0.5	0.5	0.1	0.8	0.2
Residential_Assets	2,331,195.13	2,215,685.00	11,303.00	4,988,023.00	1,515,562.25
Bank_Assets	1,034,718.12	1,080,174.00	42,534.00	1,971,164.00	551,003.19

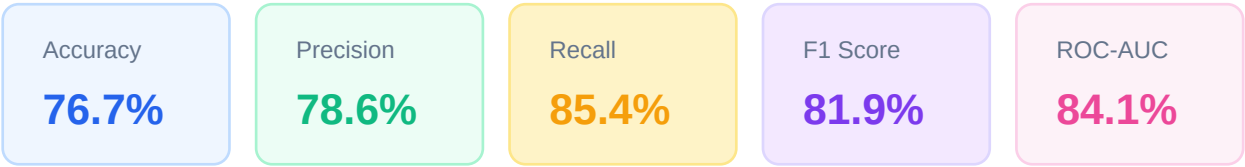
6. Outlier Analysis (IQR Method)

Feature	Outlier Count	Outlier %	Severity
No significant outliers	-	-	-

7. Distribution of Key Features

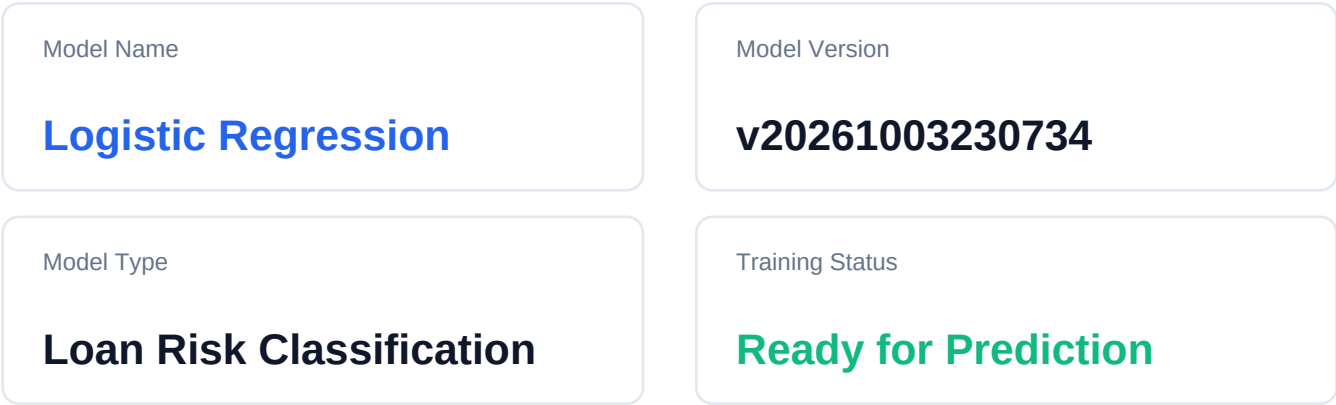


8. Model Performance



Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.767	0.786	0.854	0.819	0.841

9. Active Model Details



10. New Applicant Prediction

Applicant Information

Age	28.0
Employment Years	14.0
Annual Income	1077576.0
Coapplicant Income	38102.0
Loan Amount	798589.0
Loan Term Years	5.0
Cibil Score	493.0
Existing Loans	1.0
DTI Ratio	0.29
Residential Assets	2339767.0
Bank Assets	435423.0
Gender	Female
Married	Yes
Dependents	3+

Prediction Result

LOW RISK

Prediction Confidence: 94.0%

Approval Probability: 94.0%

Risk Probability: 6.0%

Decision: APPROVE

11. Final Summary

The system analyzed 102 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset.

Total Records

102

Total Columns

21

Selected Model

Logistic Regression

Model Accuracy

76.7%

Latest Applicant Result

LOW RISK

Approval Probability

94.0%